

ASSIGNMENT 3: PHOTO ESSAY

Customer, Technology and Data Systems Reshaping Value at KazMunayGas

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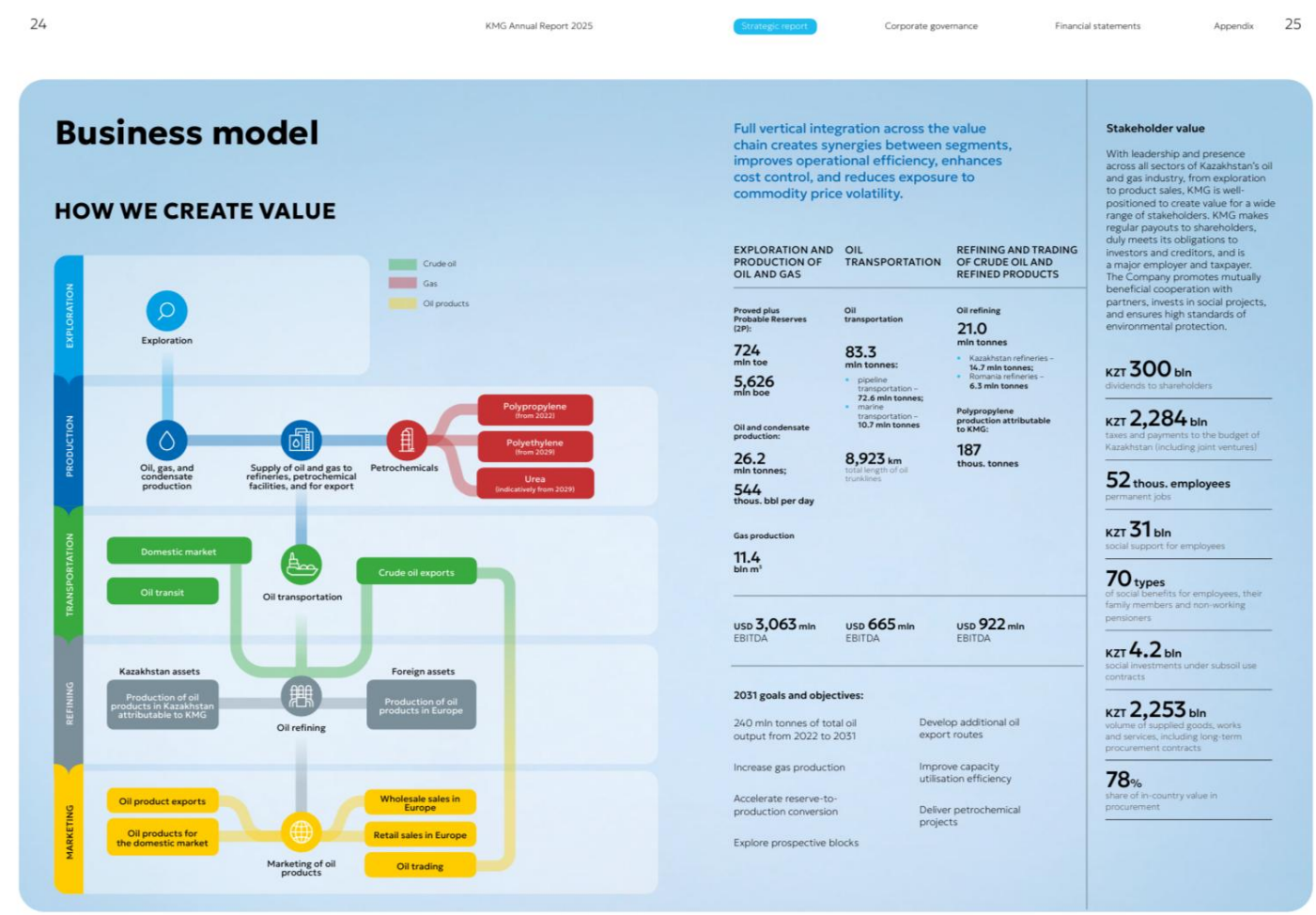
Company A: Shell

Company B: KazMunayGas

Position statement

KazMunayGas (KMG) can create durable value by combining its integrated energy value chain with carefully governed digital technologies, while adapting useful Shell practices to Kazakhstan's energy-security needs rather than copying strategies that could undermine domestic stability.

Visual 1. KazMunayGas headquarters and integrated value chain



Source: JSC NC KazMunayGas, Annual Report 2025, Strategic Review, p. 24-25 (p.14 in pdf)

Analytical commentary

KMG is Kazakhstan's national oil and gas company and operates across exploration, production, transportation, refining and retail. This integrated structure connects resources with customers. KMG's domestic energy-security role means performance must be assessed through commercial results and continuity of supply.

Visual 2. Downstream retail touchpoints and B2B fuel allocation

Export of oil products broken down by share and supply destination

Oil products	2024			2025		
	Volume, tonne	Country	Share, %	Volume, tonne	Country	Share, %
Fuel oil	346,759	Netherlands	100	295,539	Netherlands	100
Vacuum gas oil	107,552	Netherlands	100	1,137	Netherlands	100
High-purity paraxylene	51,790	China	100	32,669	China	100
Benzene	11,937	China	100	19,262	China	100
Total coke	79,866	China	79	110,360	China	94
	19,967	Azerbaijan	20	2,708	Türkiye	2
	680	Tajikistan	1	4,123	Tajikistan	4
Calcined coke	4,704	Azerbaijan	100	3,836	Türkiye	39
				464	Tajikistan	5
				5,453	China	56
Sulphur	760	Sweden	30	7,652	Europe	74
	1,806	Türkiye	70	2,757	Türkiye	26
AI-92 petrol (Shymket Refinery)				1,276	Kyrgyzstan	14
				6,825	Uzbekistan	76
				923	Afghanistan	10
Total	**625,823**			**494,983**		

Oil product imports in 2025
In accordance with the commission of Kazakhstan's Government, KMG imported 295 thous. tonnes of diesel fuel supplied to KTZ – Freight Transportation from Russian refineries.

To make up for the domestic shortage of jet fuel, Kazakhstan's Government commissioned KMG to purchase 25.1 thous. tonnes from Russian refineries.

KC Energy Group

Strategic transformation
To operate effectively at Kazakhstan's oil product market and ensure energy security, KC Energy Group was established on 27 December 2023.

Shareholder structure
KC Energy Group operates on the same principles applied to PETROSUN and has the same shareholder structure:
- CNPC International in Kazakhstan – 51%;
- KMG – 49%.

Operations
The company's business model represents a closed cycle including:
1. purchasing crude oil from affiliated oil companies;
2. oil refining at Kazakhstan refineries;
3. wholesale distribution of finished oil products in both domestic and export markets.

Energy security and social obligations
KC Energy Group prioritises seamless fuel supply to strategic industries as planned by Kazakhstan's Ministry of Energy.

In 2025, the company accounted for 46% in total refining at three Kazakhstan refineries and sold 8,021.3 thous. tonnes of oil products.

In 2025, the company accounted for 46% in total refining at three Kazakhstan refineries and sold 8,021.3 thous. tonnes of oil products. The majority of products are sold domestically, which testifies to strong demand and the company's stable positions in Kazakhstan.

- 7,468.1 thous. tonnes (93.1%) – domestic market, including 5,828.8 thous. tonnes (78.1%) of light oil products.
- 553.22 thous. tonnes (6.9%) – exports.

In 2025, as part of social obligations, KC Energy Group supplied 1,396.1 thous. tonnes of oil products:
- 360.8 thous. tonnes (25.8%) of diesel fuel to agricultural producers at prices discounted below the market as recommended by an authorised body;
- 114.2 thous. tonnes (8.2%) of fuel oil to social and production facilities at prices discounted below the market as recommended by an authorised body;
- 331.9 thous. tonnes (23.8%)of jet fuel to airports and airlines;
- 420.1 thous. tonnes (30.1%) of liquefied petroleum gas to gas distribution companies at prices recommended by an authorised body;
- 163.9 thous. tonnes (11.7%) of road bitumen from Pavlodar Refinery to local road construction companies at prices discounted below the market as recommended by an authorised body;
- 4,965 thous. tonnes (0.4%) of fuel oil as humanitarian aid to Tajikistan.

In 2025, 10% of all AI-92/AI-95 petrol, diesel and jet fuel were sold at electronic trading platforms, with total volume of such sales reaching 398.9 thous. tonnes. Despite volatile exchange prices in 2025, KC Energy Group earned extra income of KZT 3,321 mln including VAT, jet and diesel fuel being the largest contributors to such income from exchange trading.

The company put in a lot of effort to sell 534.3 thous. tonnes of AI-92 petrol and diesel fuel from regional oil depots which helped prevent facility stoppage at times of low demand and provide regional filling stations with socially important oil products when demand was high.

Amid an unprecedented drop in domestic demand for AI-92 petrol in the first half of 2025, a quick and smooth action from KC Energy Group, KMG's Corporate Centre and an authorised body paved the way to the sanctioned exports of 50.2 thous. tonnes of surplus AI-92 petrol in April–May 2025, stabilising the situation in the domestic market.

KMG International

Refining volumes at KMG International assets (Petromidia and Vega refineries) increased by hefty 27.9% to 6,282 thous. tonnes due to production recovery after the scheduled overhaul and the restart of the mild hydrocracker in 2024.

Hydrocarbon refining volumes, thous. tonnes

Refinery	2023	2024	2025
Petromidia Refinery	5,012	4,619	5,886
Vega Refinery	374	293	395
Total	**5,387**	**4,912**	**6,282**

In January 2025, meetings were held with the Azerbaijan Caspian Shipping Company (ASCO), the Port of Baku, ADY Express (a subsidiary of Azerbaijan Railways), EVRASCON (a designer of ferry terminals), Georgian Railways, and the TRACECA Secretariat to discuss the project and further steps. With all the parties demonstrating interest in and pledging their support to the project; an agreement was reached to jointly coordinate further steps in the project. In March 2025, the project was presented to members of the TMTM Association receiving its full support.

On 1 July 2025, KMTF, ADP, ASCO and CIMS signed a memorandum of cooperation on advancing Trans-Caspian ferry transportation between the ports of Kuryk and Alyat. The parties are also working to select a consultant for the feasibility study aimed at determining the potential cargo base, infrastructure needs, investment volume and further project format. At this stage, Kazakhstan Temir Zholy was invited to participate in the project. In addition, the World Bank's International Finance Corporation (IFC) expressed interest in co-financing the feasibility study. The study will be finalised in 1H 2026 and followed by a decision on further implementation.

Plans for 2026

1. **Open-sea fleet expansion:**
   completing the acquisition of two Aframax or Suezmax tankers.
2. **Cooperation with AD Ports Group:**
   - implementing the project to build a 780 TEU[1] container vessel;
   - joint design of a feasibility study of a Trans-Caspian ferry system involving ASCO (Azerbaijan) and Kazakhstan Railways.



Downstream

Sales of KMG-produced oil and condensate, thous. tonnes

Assets	2023			2024			2025		
	Export	Domestic market	Total	Export	Domestic market	Total	Export	Domestic market	Total
Operating assets[2]	5,406	8,302	13,708	5,884	8,518	14,402	5,631	8,740	14,371
including subsidiaries and associates[3]	2,955	4,980	7,935	3,169	5,224	8,383	2,985	5,411	8,396
Megaprojects[4]	9,978	9	9,987	9,616	1.04	9,987	11,908	1.04	11,909
Total	**15,384**	**8,311**	**23,695**	**15,500**	**8,519**	**24,019**	**17,539**	**8,741**	**26,281**

[1] Twenty-foot equivalent unit.
[2] Ozenmunaigas, Embamunaigas, Karazhanbasmunai, Kazgermunai, PetroKazakhstan Inc., Kazakhturkmunay, Kazakhoil Aktobe, Mangistaumunaigaz, Urikhtau Operating, Dunga, Ural Oil and Gas.
[3] Ozenmunaigas, Embamunaigas, Kazakhturkmunay, Urikhtau Operating.
[4] KMG Kashagan, KMG Karachaganak, Tengizchevroil.

Pursuant to the President's commission and further commissions from the Government of the Republic of Kazakhstan, KMG continues to introduce the combined scheme at its refineries, taking into account the amendments being made to the Law of the Republic of Kazakhstan On State Regulation of Production and Sales of Certain Oil Products with respect to oil supplies to local refineries. The combined scheme and amendments mentioned above provide for a subsoil user's affiliate to be recognised as an oil supplier subject to at least one of the following conditions:
- the oil supplier owns at least 50% of shares in the subsoil user;
- the subsoil user owns at least 50% of shares in the oil supplier;
- the oil supplier and subsoil user are controlled by one or more persons owning, directly or indirectly, a total of at least 50% of shares in the oil supplier and subsoil user.

Ozenmunaigas, Embamunaigas, Kazakhturkmunay and Urikhtau Operating supply Atyrau, Pavlodar and Shymkent refineries with KMG's own crude oil, and the resulting refined products are subsequently sold wholesale domestically or for export. In 2025, sales of own oil and gas condensate to meet domestic demand amounted to 8,741 thous. tonnes (8,519 thous. tonnes in 2024), including 5,411 thous. tonnes of crude oil (5,224 thous. tonnes in 2024) supplied from operating assets – Ozenmunaigas, Embamunaigas, Kazakhturkmunay, Urikhtau Operating – to Atyrau, Pavlodar and Shymkent refineries for further refining and oil product sales.

Refining assets

KMG's hydrocarbon refining operations embrace four largest refineries in Kazakhstan and the sole asset in the EU – two refineries in Romania producing a wide range of products.

KMG refineries

Indicator	Kazakhstan				Romania	
	Atyrau Refinery	Pavlodar Refinery	Shymkent Refinery	Caspi Bitum	Petromidia Refinery	Vega Refinery
Location	Atyrau	Pavlodar	Shymkent	Aktau	Năvodari	Ploiești
Commissioning date	1945	1978	1985	2013	1979	1905
Design refining capacity, mln tonnes	5.5	6.0	6.0	1.5	6.0[1]	0.5
Refinery utilisation rate in 2025, %	99	96	104	46	98[2]	79
KMG interest, %	99.53	100	49.72	50	54.6	54.6
Nelson Index	13.9	10.5	8.2	–	10.5	–
Energy Intensity Index (EII)	114	113	113.3	–	92.5	–
Refinery co-owners	–	–	CNPC	CITIC	Romanian Government	Romanian Government

[1] Design capacity includes refining 5 mln tonnes of crude oil and 1 mln tonnes of other hydrocarbons per year.
[2] Petromidia Refinery utilisation rate is 97.5% based on Solomon Associates' methodology.

Source: JSC NC KazMunayGas, Annual Report 2025, Strategic Review, p. 104-113 (p.54-58 in pdf)

Analytical commentary

KMG serves business customers and individual motorists through different channels. Businesses may need planned volumes, dependable delivery and clear procurement terms, while retail customers value access, availability and consistent service. Channel strategy should reflect these different needs.

Visual 3. Marketing mix (5 Ps) for KMG domestic fuel products

P	Decision	Reason
Product	Euro-5 compliant diesel with depressant-dispersant additives for winter.	Prevents fuel gelling down to -30°C in northern KZ regions; matches environmental standards.
Price	Leader pricing under Ministry of Energy RK state price cap orders.	Order No. 171 regulates domestic fuel caps, preserving affordability and curbing inflation.
Place	Direct refinery rail loading terminals + 300+ retail gas station network.	Ensures primary transport corridors and agricultural hubs receive priority fuel supply.
Promotion	Direct B2B long-term contracts, digital fuel cards, national energy messaging.	Secures guaranteed B2B offtake while reinforcing KMG's sovereign brand trust.
People	Certified rail loading operators & station staff trained under TUMAR AI safety.	Guarantees zero-harm industrial safety and professional bilingual customer service.

Item	Your answer
Life-cycle stage of the product, with reason	Maturity. The petroleum products market is mature, requiring high operational efficiency and customer retention.
Pricing strategy (skimming, penetration, leader, bundling, odd-even, prestige)	Penetration / Leader bundling in the context of government price regulation.

Source: Weekly record 6, step 3.

Analytical commentary

KMG's marketing mix focuses on reliable fuel supply, affordability, and nationwide distribution, using regulated leader pricing, B2B contracts, and trained personnel to support Kazakhstan's energy security. Its mature product market requires operational efficiency and customer retention rather than aggressive market-entry pricing.

Visual 4. Five-stage journey for commercial fuel procurement

Five stages of the buying process. Cultural, social, individual and psychological factors affect every step.

Stage	Customer thinks or does	Real touchpoint	Helping P	Pain point
1 Need recognition	The fleet manager is planning to purchase diesel before sowing/harvesting.	B2B portal / Regional sales department	Product	Fear of freezing fuel in winter or a shortage in the region.
2 Information search	Compares tariffs, volume discounts, and filing schedules.	KMG B2B web portal and tender platforms	Price	Non-transparent quotes or delays from private intermediaries
3 Evaluation of alternatives	Verifies the Euro-5 quality certificates and payment deferral terms.	Quality certificates from refinery laboratories	Place	Requirements: 100% prepayment or distance to loading ramps.
4 Purchase decision	Sign an annual contract or scans a smart card at the overpass.	B2B portal / RFID card	People	Queues and delays in processing at railway overpasses.

Stage	Customer thinks or does	Real touchpoint	Helping P	Pain point
5 Post-purchase behavior	It analyzes fuel consumption via GPS and requests reconciliation statements.	KMG B2B mobile app	Promotion	Discrepancies between meter data and invoices.

Source: Journey framework adapted from OpenStax, Introduction to Business, 2nd ed., Ch. 11, Weekly record 6, step 4.

Analytical commentary

KMG's customer journey focuses on helping fleet managers secure reliable fuel supplies through B2B portals, quality certification, contracts, and digital tracking tools. However, prepayment requirements, loading delays, and invoice discrepancies may reduce customer satisfaction and weaken long-term loyalty.

Visual 5. Human-only versus AI-supported marketing plan

Criterion	Human 0–4	AI 0–4	Evidence
Theory fit	4	3	Human plan strictly separates B2B bulk supply and B2C retail 5P frameworks, AI blended wholesale and retail channels together.
Insight	3	2	Human plan incorporates seasonal winter diesel additive requirements (−30°C), AI generated generic, non-regional marketing advice.
Accuracy	4	1	Human plan grounds pricing in state regulations, AI hallucinated illegal dynamic surge pricing during peak hours.
Brand consistency	4	3	Human plan reflects KMG's sovereign mandate and social responsibilities; AI drafted generic corporate marketing copy.
Ethics	4	3	Human plan ensures compliance with RK antitrust and consumer laws, AI ignored local regulatory frameworks.
Kazakhstan context	4	1	Human plan enforces bilingual Kazakh/Russian interfaces and RK data residency; AI assumed US/EU e-commerce norms.
Total (max 24)	23	13	Human plan significantly outperforms AI due to legal accuracy and Kazakhstan market fit.

Source: Weekly record 6, step 9

Analytical commentary

The comparison suggests AI can organise marketing options, but its output needs review for accuracy, brand fit, ethics and Kazakhstan-specific context. The matrix totals Human 23/24 and AI 13/24 if the recorded scores are retained. These scores reflect this rubric, not AI performance universally.

Visual 6. AI prompt and verification log: KEEP / FIX / DROP

AI statement	Check	Evidence or source	Verdict
Implement dynamic surge pricing on Euro-5 fuels during peak hours.	Facts and Local	Order of the Minister of Energy of RK No. 171 on state-regulated fuel prices.	☐ Keep ☐ Fix ☒ Drop
Integrate B2B IoT API telemetry into the customer portal for real-time fleet fuel monitoring.	Theory and Brand	KMG Digital Transformation Roadmap 2025-2027 & ABAI Platform brief	☒ Keep ☐ Fix ☐ Drop
Deploy an English-only AI customer support chatbot for B2C drivers.	Local and Ethics	Law of RK No. 274-IV "On Consumer Rights Protection" (Art. 24).	☐ Keep ☒ Fix ☐ Drop
Transition 100% of fuel sales exclusively to app-based e-commerce pre-ordering.	Facts and Local	Physical reality of refinery loading racks (ANPZ/PNHZ) and retail network.	☐ Keep ☐ Fix ☒ Drop
Introduce a Telegram-based loyalty stamp card for B2C retail customers.	Theory and Brand	Qazaq Oil / KMG Retail marketing practices	☒ Keep ☐ Fix ☐ Drop
Claim 99.9% carbon-neutral fuel delivery across all domestic routes.	Facts and Ethics	KMG 2025 Annual Report (Scope 1 & 2 emissions data, p. 112).	☐ Keep ☐ Fix ☒ Drop

Source: Weekly record 6, step 8

Analytical commentary

The AI review shows that KMG should adopt practical digital improvements, such as B2B fuel monitoring and a Telegram loyalty card, while rejecting surge pricing, fully app-based sales, and unsupported carbon-neutrality claims. An English-only chatbot should be revised to support Kazakhstan’s multilingual customers and local needs.

Visual 7. Five-stage AI/data workflow for refinery maintenance

Step 5: the five-stage workflow

Stage	What happens	Data or tool	Who is responsible	Time or frequency
1 Input	IoT sensors record real-time vibration and temperature data on pumps and compressors at ANPZ	IoT vibration and thermal sensors, SCADA telemetry	Shift Operator (*Shift Operations Team*)	Continuous (24/7, 1 Hz frequency)
2 Analysis	ABAI platform ML algorithms process streaming telemetry to detect micro-friction anomalies 14 days prior to potential failure	ABAI Platform Machine Learning algorithms (KMG Engineering)	Lead Data Scientist / Predictive Analytics Module	Automated every 5 minutes
3 Human verification	Chief Mechanical Engineer analyzes the AI alert, inspects live telemetry, and conducts an on-site physical inspection to rule out false positives	ABAI Dashboard, visual inspection, portable vibrometer	Chief Mechanical Engineer (ANPZ)	Within 15 minutes of AI alert
4 Decision	Based on verified telemetry, the engineer approves a preventive maintenance work order in SAP ERP	SAP ERP Asset Management System	Chief Mechanical Engineer / Maintenance Workshop Head	Within 1 hour
5 Monitoring	Tracking post-repair vibration metrics, evaluating Mean Time Between Failures (MTBF), and updating weekly KMG executive KPI dashboards.	KMG MIS Dashboard, plant throughput reports	Plant Operations Director (ANPZ)	Weekly / Monthly

Source: Weekly record 7 workflow artifact.

Analytical commentary

The proposed workflow uses AI-powered predictive maintenance to detect equipment anomalies, verify alerts through human inspection, and schedule preventive repairs through SAP ERP. Continuous monitoring of equipment performance can reduce unexpected failures, improve operational reliability, and support data-driven maintenance decisions.

Visual 8. TUMAR system

Fire safety
No fires occurred at KMG's facilities in 2025.

To strengthen fire safety at its subsidiaries and associates, KMG developed and approved the 2025–2027 HSE Improvement Roadmap. Fire prevention activities were carried out throughout the year under the Roadmap, along with scheduled fire drills at all subsidiaries and associates.

Pursuant to a quadripartite Memorandum of Cooperation on Emergency Prevention and Response signed by the State Oil Company of Azerbaijan, KMG, Tatneft, and Uzbekneftegaz, drills were held on 29 August 2025 to practise responding to a well fire with casualties at Embamunaigas. The drills involved personnel and resources from the State Fire Service of the Atyrau Region Emergency Situations Department under Kazakhstan's Ministry of Emergency Situations as well as delegations from foreign countries.

Additional drills were conducted to practise extinguishing a simulated fire in a vertical steel oil storage tank at Caspi Bitum, with participation from Samruk-Kazyna representatives and the Mangistau Region Emergency Situations Department.

KMG's Corporate Centre held two fire drills in 2025, involving evacuation and simulated building fire suppression.

Throughout the year, subsidiaries and associates received methodological support and guidance on fire safety management via letters addressing enhanced fire protection measures during fire-hazardous, heating, and holiday periods, along with clarifications of fire safety regulations.

HSE interaction with contractors
In 2025, as part of the 2025–2027 HSE Improvement Roadmap, KMG Group implemented a series of measures to strengthen HSE performance among its contractor and subcontractor organisations. These measures were informed by incident analysis in the Upstream and Oilfield Services business segments, which carry elevated operational risks.

Key priorities included clarifying responsibility allocation, establishing measurable KPIs, classifying work by risk level (low, medium, high), and introducing safety-specific KPIs for contractors.

To strengthen oversight over engagement with contractors, mechanisms were developed to enhance the role of contract administrators, line managers, and HSE departments in:

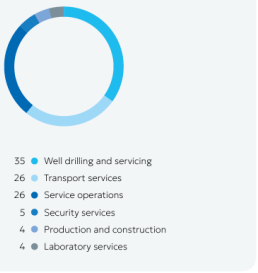
- conducting internal audits of contractor and subcontractors;
- analysing root causes of incidents and non-conformities;
- developing and monitoring corrective actions.

At the November 2025 HSE Forum – attended by contractor leadership, subsidiary and associate CEOs, and contract administrators from the Upstream and Service business segments – key management and communication priorities were defined to foster a partnership approach to industrial safety and increase contractors' accountability for their performance.

During the year, KMG's Corporate Centre audited 18 subsidiaries and associates to assess their HSE engagement processes with contractors, issuing 824 recommendations for improvement at both contractor organisations and the subsidiaries and associates themselves. A total of 5,883 industrial safety inspections of contractor operations at KMG subsidiaries, associates, and branches were also conducted.

An assessment of contractor HSE performance in 2025 showed that industrial incidents were most frequent in well drilling and well servicing (35%), followed by service operations (26%) and transport services (26%).

Contractor incidents by type of service provided, %



Contractor injury, fatality, and motor vehicle crash rates

Indicator	2023	2024	2025	IOGP[3]
Lost Time Incident Rate (LTIR), per mln person-hours	0.36	0.32	0.45	0.24
Fatal Accident Rate (FAR), per 100 mln person-hours	12.0	3.86	10.38	0.84
Motor Vehicle Crash Rate (MVCR), per mln km travelled	0.28	0.05	0.07	0.03

Digitisation of HSE processes
In 2025, KMG continued developing TUMAR, an automated intelligent industrial safety system for preventing incidents and reducing the negative impact on employees when carrying out hazardous operations, including underground well servicing.

A pilot launch of the TUMAR automated information system was conducted in January 2025 across ten well workover and servicing crews at the Zhetybai and Kalamkas fields (operated by Mangistaumunaigaz) within Oil Services Company. All three modules operated as intended during the pilot.

Work is now underway to deploy the system to 60 well workover and servicing crews at the same fields within Oil Services Company.

In 2026, the project is scheduled for rollout at Ozenmunaigas, OzenMunayService, Karazhanbasmunai, and Embamunaigas.

Occupational health and hygiene
KMG manages occupational health and hygiene in line with Kazakhstan's laws and international standards governing occupational health, hygiene, and industrial sanitation. The Company pursues an active policy to protect and improve the health of its employees involved across all production processes,

In 2025, non-work related fatalities among employees of KMG's subsidiaries and associates decreased by four incidents, or 33%, compared to 2024. Cardiovascular disease remained the primary cause.

Source: JSC NC KazMunayGas, Annual Report 2025, Digitalization section, p. 192

Analytical commentary

KMG's TUMAR system digitises HSE processes to prevent industrial incidents, with a pilot successfully conducted in January 2025 across 10 well-servicing crews at the Zhetybai and Kalamkas fields. Following the pilot, KMG planned expansion to 60 crews, with further rollouts scheduled for additional subsidiaries in 2026.

Visual 9. Governance risk register: risks and controls

Risk	What could go wrong	Control	Owner	Trigger
Privacy	Leak of operational refinery logs or engineer personal data when exporting to external cloudы	Access restriction, encryption over secure KMG VPN, local server hosting at Astana Data Center	IT Security Lead	Unauthorized export attempt / Annual access audit
Risk	**What could go wrong**	**Control**	**Owner**	**Trigger**
Security	Cyberattack, ransomware, or unauthorized access to ANPZ SCADA industrial control networks	Network air-gapping for OT network, mandatory 2FA, daily offsite encrypted backups	Chief Information Officer (CIO) / Cybersecurity Officer	Failed login attempt from unauthorized IP / Compromised password
Bias	ML model trained on summer telemetry misinterprets equipment wear at sub-zero winter temperatures (–30°C)	Weekly algorithm recalibration using 5-year seasonal temperature datasets	Lead Data Scientist	Prediction accuracy deviation >5% during winter season
Reliability	AI model misses critical bearing wear or triggers a false alarm recommending an unnecessary shutdown of a healthy unit.	Mandatory Human Review Point: Chief Mechanical Engineer physical inspection and sign-off required before any shutdown	Chief Mechanical Engineer (ANPZ)	Discrepancy >10% between AI prediction and manual vibrometer reading
Accountability	Engineers excuse bad decisions by claiming "the AI told me to do it," abdicating human responsibility	Regulatory policy designating AI as Decision Support System only; mandatory personal engineer sign-off on all work orders	Plant Operations Director (ANPZ)	Work order execution without engineer sign-off
Escalation	AI registers rapid multi-pump vibration spikes simultaneously across multiple units	Escalation protocol: automated alert to Chief Mechanical Engineer (within 15 min) and Plant Chief Engineer (within 1 hr)	Senior Duty Engineer	3 or more critical anomaly flags within 1 hour
Fallback	Total IoT sensor network outage, AI server crash, or power loss to control network	Manual Fallback: Immediate transition to 2-hour physical operator walks with paper logbooks and analog pressure gauges	Shift Supervisor	AI/SCADA server unavailability exceeding 15 minutes

Source: Weekly record 7

Analytical commentary

KMG's AI-supported maintenance workflow requires strong governance to manage privacy, cybersecurity, model bias, and system reliability risks. Human sign-off, escalation procedures, and manual fallback processes help maintain accountability and operational safety if the AI system fails or produces inaccurate predictions.

Visual 10. Strategic transfer matrix: Shell to KazMunayGas

Comparative Analysis and Transfer Judgments

Directly adopting Shell's strategy would compromise KMG's national mandate [1]. Transfer of technologies and managerial practices requires a structured, nuanced framework with explicit transfer judgments:

1. Adopt: AI Predictive Maintenance & Computer Vision Safety Systems
Reasoning: KMG should fully adopt Shell's practice of enterprise-wide predictive failure analytics and CCTV computer vision safety monitoring (modeled on Shell's T-Pulse). Extending TUMAR and ABAI across all domestic refineries (ANPZ, PNHZ, PKOP) will eliminate unplanned downtime and drastically reduce industrial accidents [3], [4].

2. Adapt: Climate Strategy & Decarbonization Framework
Reasoning: Shell's Net-Zero framework must be adapted to Kazakhstan's fossil-reliant economy. KMG should focus on Scope 1&2 methane abatement (OGMP 2.0 Gold Standard), hybrid renewable/gas generation (Zhanaozen 247 MW), and CCUS screening at gas processing units rather than divesting producing hydrocarbon assets [3], [4], [5].

3. Pilot: Digital Twin Reservoirs & Advanced EOR / Slim Hole Drilling
Reasoning: Shell's digital twin subsoil modeling and advanced EOR techniques should be piloted across declining mature fields (Embamunaygas, Ozenmunaygas) to optimize sweep efficiency and extend field life [3], [4].

4. Delay: Mass Capital Allocation to Green Hydrogen & EV Retail Infrastructure
Reasoning: Unlike Shell (which expands Shell Recharge and green hydrogen electrolysers), KMG should delay heavy CAPEX in green hydrogen and EV charging networks until EV fleet adoption and market demand mature in Central Asia, restricting current involvement to R&D desk studies [3], [4], [5].

5. Reject: Pure Market Pricing & Aggressive Divestment of Social Downstream Assets
Reasoning: Shell's strategy of shutting down or selling lower-margin refining assets must be explicitly rejected. Abandoning domestic fuel subsidies or selling domestic refineries would cause severe fuel shortages, economic inflation, and social disruption in Kazakhstan [1], [3], [5], [9].

Source: Shell plc, Annual Report and Accounts 2025; KMG Annual Report 2025

Analytical commentary

Shell is comparative evidence, not a template KMG should copy wholesale. KMG may consider predictive maintenance, adapt decarbonisation to local priorities and pilot digital twins where feasible. Rejecting asset divestment or unrestricted price floating requires evidence about KMG’s mandate, Kazakhstan’s rules and energy security.

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